

Tokenization = In LM

Breaking raw input (text) into smaller pieces called

Tokens.

Text $\rightarrow$ Codingdidi is a girl!

Tokens :-

["Coding"] ["didi"] ["is"] ["a"] ["girl"] ["."]

Model don't read text

They read TOKENS $\rightarrow$

converted to numbers $\rightarrow$

embeddings

Why Tokenization?

- ① Convert text $\rightarrow$ manageable unit
- ② Handle unknown words
- ③ Reduce memory usage
- ④ Enable generalization

Notice:

- $\rightarrow$ Spaces Matter
- $\rightarrow$ Word may be split.
- $\rightarrow$ Tokens are not always full words.

Types of Tokenization

① Word Tokenization

↓
splits by words.

"I Love AI"

[I] [love] [A]

- o→ Huge Vocabulary
- o→ Unknown words break system
- o→ Doesn't handle Typos/ slang well.

② Character Tokenization

split by characters $\Downarrow$

"I Love AI" $\Rightarrow$

[ɪ] [ɪ] [o] [v] [e]

Pros

↳ No unknown words.

Cons

- ↳ Very long sequences
- ↳ Harder to learn meaning.

③ Subword Tokenization

↓
split into meaningful
chunks.

"Unhappiness"

[uni] [happy] [ness]

- Handles unknown words.
- keeps Vocabulary manageable.
- preserves meaning.



- Subword Tokenization is used in modern LMs (BPE, Word Piece, sentence-piece etc.)

Popular Algorithm.

1. BPE (Byte pair Encoding)

- o starts with characters
- o Merge most frequent pairs
- o Used in early GPT → GPT-2

2. Word Piece.

- o Build Tokens based on probability.
- o keeps most useful subword.
- o Example — playing — [Play] [ing]
- o Used in BERT

3. Sentence piece.

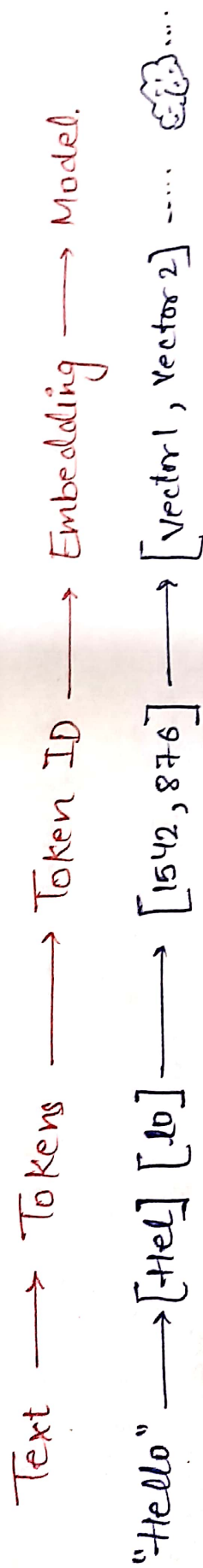
- o Treats Text as raw stream
- o Doesn't rely on spaces.
- o Used in T5, LLaMa.

4.

Unigram Language Model.

- o chooses most probable token set.
- o Most flexible than BPE

What happens after Tokenization



** Tokenization affects everything

Context length

Models don't see words.

they see Tokens.

"1000 words $\neq$ 1000 Tokens

1 token $\approx$ 0.75 word.

Performance.

Bad Tokenization

$\rightarrow$ breaks meaning

$\rightarrow$ hurts accuracy.

Good Tokenization

$\rightarrow$ better understanding

$\rightarrow$ better generation

Cost.

APIs charge per token.

More Token =

More Cost